Bow Valley College

Course Outline

TECH9211

Enterprise Networking, Security, and Automation
Spring 2022 - Fall 2022

Last Updated: 4/26/2022 1:57:26 PM

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TECH9211 Enterprise Networking, Security, and Automation

COURSE DESCRIPTION

The Enterprise Networking, Security, and Automation course describes the architectures and considerations related to designing, securing, operating, and troubleshooting enterprise networks. This course covers wide area network (WAN) technologies and quality of service (QoS) mechanisms used for secure remote access. It also introduces software-defined networking, virtualization, and automation concepts that support the digitalization of networks. Students gain skills to configure and troubleshoot enterprise networks and learn to identify and protect against cybersecurity threats. They are introduced to network management tools and learn key concepts of software-defined networking, including controller-based architectures and how application programming interfaces (APIs) enable network automation.

REQUISITES	None
EQUIVALENTS	None
CREDITS	0
HOURS	128
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Configure single-area Open Shortest Path First (OSPF)v2 in both point-to-point and multiaccess networks.	2, 5
2	Explain how to mitigate threats and enhance network security using Access Control Lists (ACLs) and security best practices.	1, 2, 4, 5
3	Implement standard IPv4 Access Control List (ACLs) to filter traffic and secure administrative access.	1, 4, 5
4	Configure Network Address Translation (NAT) services on the edge router to provide IPv4 address scalability.	2, 4, 5, 6
5	Explain techniques to provide address scalability and secure remote access for WANs.	1, 2, 4
6	Explain how to optimize, monitor, and troubleshoot scalable network architectures.	1, 2, 4, 5
7	Explain how networking devices implement protocols to manage the network and quality of service (QoS).	1, 2, 5
8	Explain how technologies such as virtualization, software defined networking, and automation affect evolving networks.	1, 2, 5

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Single-Area OSPFv2 concepts, Single-Area OSPFv2 configuration, network security concepts, ACL concepts
Week 2	ACLs for IPv4 configuration, NAT for IPv4, WAN concepts, VPN and Internet Protocol Security (IPsec) concepts
Week 3	QoS concepts, network management, network design, network troubleshooting
Week 4	Network virtualization, network automation, final assessments

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1, 2, 3, 4, 5, 6, 7, 8	Practical application	50%
1, 2, 3, 4, 5, 6, 7, 8	Final exam	50%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A grade of P is required to pass this course. Learners may be required to pass this course in order to progress in the program or to meet specific program completion requirements.

To receive a course overall pass, the learner must pass all assessments.

Please consult with the program area or the Bow Valley College website for further details.

GRADING SCHEME

Symbol	**Description**	**Value in GPA**
P	Pass	N/A
F	Fail	N/A
BD	Basic/Developing (Foundational programming courses only)	N/A

REQUIRED LEARNING RESOURCES

Computer requirements:

Standards	**Hardware**	***Software***
Processor	i7	
RAM (memory)	16 GB RAM or greater	Windows 10 Pro/education 64-bit
Hard drive & storage	512 GB SSD or greater	
Video card	On-board integrated	(MacOS is not supported)
Screen size	15" or greater	

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online

proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.